

Reflective Journal – Lab 05 Data Preparation

Lab 05 was the first time I really felt the importance of data preparation in a machine learning workflow. Before this module, I understood the idea of training models and making predictions, but I didn't realize how much of the work happens before you even get to the modeling stage. Working with the Titanic dataset made that clear. The data wasn't perfect, and dealing with missing values, categorical variables, and inconsistent formats showed me how essential it is to prepare the dataset properly.

One of the biggest things I learned was why the **median** is often better for imputing missing values than the mean. The Titanic dataset has a wide range of ages and some extreme fare values. Using the mean could pull the imputed values in a direction that doesn't represent the typical passenger. The median is more stable and less sensitive to outliers, which makes it a better choice for filling in missing ages. Seeing how this small decision affects the dataset helped me understand how preprocessing choices can influence the model later.

Another important part of the workflow was **one-hot encoding**. Machine learning models can't understand text, so converting categorical variables like *Sex* and *Embarked* into numerical columns is necessary. I didn't realize how much this step matters until I saw how the encoded DataFrame changed. Dropping the first category to avoid multicollinearity also showed me that preprocessing isn't just about converting data — it's about making sure the model can interpret it correctly without being confused by redundant information.

Feature scaling was another step that helped me understand workflow complexity. Some models rely on distance calculations or gradient-based optimization, and those models can be affected by differences in scale. Even though decision trees don't need scaling, other models do, so scaling *Age* and *Fare* prepares the dataset for multiple algorithms. This made me appreciate how preprocessing keeps your options open and ensures the dataset is ready for whatever model you choose later.

Overall, Lab 05 helped me see how all the pieces fit together in a machine learning workflow. It wasn't just about writing code — it was about understanding why each step matters and how it affects the next part of the process. Even though this is a collaborative project, each of us had to complete our own notebook, and that pushed me to really understand the workflow instead of relying on someone else's work. Talking with my teammates and comparing our approaches also helped me see different ways to solve the same problems, which made the collaboration feel more like a real data team environment.

This lab showed me that machine learning is not just about algorithms. It's about preparing data carefully, making thoughtful decisions at each step, and understanding how those decisions shape the final model. Lab 05 gave me a clearer picture of the workflow complexity and helped me appreciate how important data preparation is in any machine learning project.